Task 8: Session 8 (AIML) - Assignment Questions

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1. Creating Sets

Write a program to create the following sets:

- A set with 5 integers.
- A set with mixed data types (integer, string, float).
- An empty set using the correct method.
- A set from the string "hello" using the set() constructor.

Print each set and its type. Add comments explaining why sets automatically remove duplicates.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Integer Set: {50, 20, 40, 10, 30}
Type: <class 'set'>
Mixed Set: {True, 3.14, 'world', 42, 'hello'}
Type: <class 'set'>
Empty Set: set()
Type: <class 'set'>
Set From 'hello': {'l', 'o', 'h', 'e'}
Type: <class 'set'>
PS D:\Mountreach_ITR_2026>
```

2. Characteristics Of Sets

Create a set: numbers = {10, 20, 30, 20, 40, 10, 50)

- Print the set and observe the output.
- Demonstrate that sets are unordered by printing the set multiple times.
- Try to access the first element using indexing (numbers[0]) and explain the error in comments.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe" d:/Mountreach_I
Set (Duplicates Removed Automatically): {50, 20, 40, 10, 30}
Printing Again (Order May Vary): {50, 20, 40, 10, 30}
Printing Again (Order May Vary): {50, 20, 40, 10, 30}
Traceback (most recent call last):
  File "d:\Mountreach_ITR_2026\Tasks\Task_8\Python_Files\Characteristics_Of_Sets.py"
    print(numbers[0])
          ~~~~~~~^^^
TypeError: 'set' object is not subscriptable
PS D:\Mountreach_ITR_2026>
```

3. Membership Testing

Write a program that:

- Takes a set of 5 colors as input from the user.
- Asks the user to enter a color name.
- Checks whether the color is present in the set using in and not in.
- Prints appropriate messages.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter 5 Colors:
 Color:Yellow
 Color:Red
 Color:Blue
 Color:Green
 Color:Purple
Your Color Set: {'Yellow', 'Purple', 'Red', 'Blue', 'Green'}
Enter A Color To Search For:Blue
Blue IS In The Set.
PS D:\Mountreach_ITR_2026>
```

4. add(), remove(), discard()

Create a set: fruits = {'apple', 'banana', 'mango'}

Perform the following operations and print the set after each step:

- Add 'orange' using add().
- Add ' banana' again (observe the result).
- Remove ' mango' using remove().
- Try to remove 'grape' using discard() (no error should occur).

Output:

```
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe" "d:/Mc
Initial Set: {'apple', 'mango', 'banana'}
After Add('orange'): {'apple', 'orange', 'mango', 'banana'}
After Add('banana') Again: {'apple', 'orange', 'mango', 'banana'}
After Remove('mango'): {'apple', 'orange', 'banana'}
After discard('grape') (grape Wasn't There): {'apple', 'orange', 'banana'}
PS D:\Mountreach_ITR_2026>
```

5. pop() & clear()

Create a set: s = {100, 200, 300, 400, 500)

- Use pop() to remove and print one element.
- Print the set after popping.
- Clear the entire set using clear().
- Print the final empty set.

Add comments explaining the difference between remove(), discard(), and pop().

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Original set: {400, 100, 500, 200, 300}
Popped Element: 400
Set After pop(): {100, 500, 200, 300}
Set After clear(): set()
PS D:\Mountreach_ITR_2026>
```

6. update() & copy()

Create two sets:

set1 = {1, 2, 3)

set2 = {3, 4, 5, 6)

- Make a copy of set1.
- Update set1 with elements from set2 using update().
- Print both the original copy and the updated set.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Original Copy Of set1: {1, 2, 3}
set1 After update() With set2: {1, 2, 3, 4, 5, 6}
PS D:\Mountreach_ITR_2026>
```

7. Union & Intersection

Create two sets:

A = {1, 2, 3, 4, 5)

B = {4, 5, 6, 7, 8)

- Print the union of A and B (using both .union() and operator).
- Print the intersection of A and B (using both .intersection() and & operator).

Add comments explaining what each operation does.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  Union Using .union(): {1, 2, 3, 4, 5, 6, 7, 8}
  Union Using | Operator: {1, 2, 3, 4, 5, 6, 7, 8}
  Intersection Using .intersection(): {4, 5}
  Intersection Using & Operator: {4, 5}
○ PS D:\Mountreach_ITR_2026>
```

8. Combined Methods

Write a program that:

- Takes 6 numbers as input from the user and stores them in a set (duplicates should be removed automatically).
- Adds two more numbers using add().
- Removes one number safely using discard().
- Prints the final set and its length using len().

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter 6 Numbers (Duplicates Will Be Ignored):
Number:34
Number:56
Number:78
Number:12
Number:32
Number:19
Set After 6 Inputs: {32, 34, 12, 78, 19, 56}
Add Extra Number 1:87
Add Extra Number 2:74
After Adding Two More: {32, 34, 74, 12, 78, 19, 87, 56}
Enter A Number To Remove (discard):19
Final Set: {32, 34, 74, 12, 78, 87, 56}
Length Of Final Set: 7
PS D:\Mountreach_ITR_2026>
```

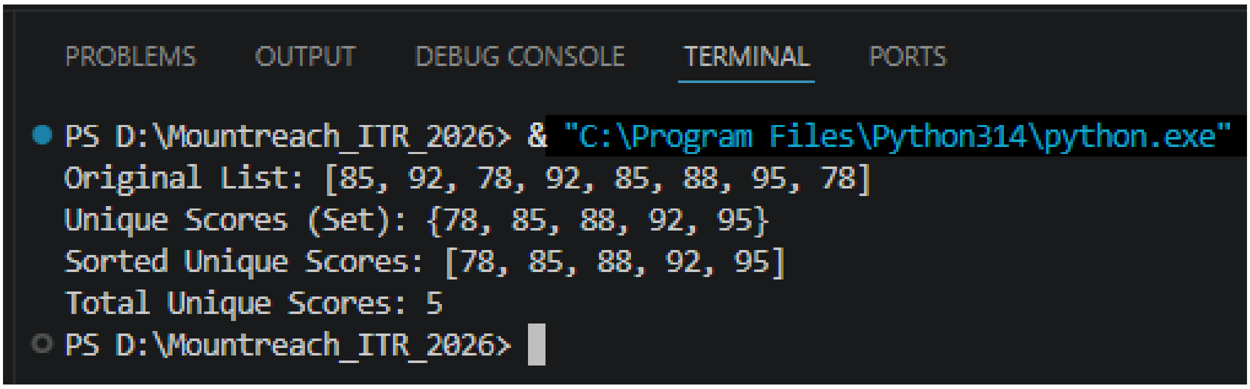
9. **Practical Application**

Write a program to remove duplicate marks from a list of student scores using a set:

scores = [85, 92, 78, 92, 85, 88, 95, 78]

- Convert the list to a set.
- Print the unique scores.
- Convert the set back to a sorted list and print it.
- Print the total number of unique scores.

Output:



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Original List: [85, 92, 78, 92, 85, 88, 95, 78]
Unique Scores (Set): {78, 85, 88, 92, 95}
Sorted Unique Scores: [78, 85, 88, 92, 95]
Total Unique Scores: 5
PS D:\Mountreach_ITR_2026>
```

10. Mini Project - Combined Concepts

Create a **Unique Item Collector** program using sets:

- Repeatedly ask the user to enter items (e.g., favorite fruits, colors, or numbers).
- Store all entered items in a set (duplicates should be ignored).
- Provide a menu with options:

  a) Add item
  b) Remove item (use discard)
  c) Show all unique items
  d) Check if an item exists
  e) Clear all items
  f) Exit

Use a while loop for the menu and appropriate set methods. Add meaningful comments.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

*** Menu ***
1. Add Item
2. Remove Item
3. Show All Unique Items
4. Check If An Item Exists
5. Clear All Items
6. Exit
Choose An Option (1-6):1
Enter Item To Add:Mihir
Mihir added. Current items: {'Mihir'}

*** Menu ***
1. Add Item
2. Remove Item
3. Show All Unique Items
4. Check If An Item Exists
5. Clear All Items
6. Exit
Choose An Option (1-6):6
Goodbye!
PS D:\Mountreach_ITR_2026>
```